Team Number:	apmcm25304033
Problem Chosen:	C

2025 APMCM summary sheet

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Contents

I. Introduction

1.1 Background

Since the 2020s, driven by the "America First" trade ideology, U.S. tariff policy has undergone major adjustments. On April 2, 2025, the Trump administration introduced the "Reciprocal Tariffs" policy, establishing a 10% baseline tariff on all trading partners and higher tariffs on around 60 countries with which the U.S. ran trade deficits, including China. The stated goal was to reduce trade deficits and promote manufacturing reshoring, but the policy violated WTO principles, including Most-Favored-Nation (MFN) treatment and tariff preferences for developing countries.

The U.S. trade-weighted average tariff rose sharply from 2.44% at the start of 2025 to 20.11% by August 2025, reaching the highest effective level since 1933 (GACC[1]) . While tariffs can protect domestic industries and generate revenue, excessively high tariffs harm exports and raise domestic consumer prices. For example, after the policy's implementation, U.S. imports from China fell 35% and exports to China declined 18% year-on-year by May 2025, narrowing the trade deficit from $30.8 billion to $18.0 billion but negatively impacting U.S. industries.

Overall, the "Reciprocal Tariffs" policy reflects a shift toward protectionism, disrupts global trade rules, and may hinder long-term global economic growth. This study aims to evaluate the policy's effects on international trade, industrial development, and consumer welfare, providing a framework to assess unilateral tariff measures and inform strategic responses to preserve the multilateral trading system.

1.2 Question Restatement

By utilizing the attached U.S. foreign trade data (2020 onwards) and supplementary information collected from authoritative channels such as WTO reports and national customs statistics, we will conduct an in-depth analysis of how U.S. tariff adjustments reshape cross-border trade patterns and

[1] [GACC, 2025] General Administration of Customs of the People's Republic of China, UN Comtrade, and USDA FAS (October 2025)

affect the economic operations of key regions. Combined with the current global trade order and the policy responses of major economies, we will put forward practical countermeasures and development suggestions for industries, such as soybean, automobile, and semiconductor, impacted by tariff changes.

1.2.1 *Question 1*

The U.S., Brazil and Argentina are top soybean exporters, and China the largest importer. Analyze their current soybean trade, build a model to assess U.S. tariff impacts on their soybean industries, and estimate post-adjustment export volume and value distribution.

1.2.2 *Question 2*

The U.S. is the second-largest auto market and top importer; 46% of its 2024 car sales were imports, mainly from Japan. Japan also produces in the U.S. or Mexico for U.S. export. Analyze Japan's U.S. auto position, build a model with Japan's non-tariff responses to study U.S. tariff impacts on their auto trade, U.S. import structure and U.S. auto industry.

1.2.3 *Question 3*

The U.S. leads the global semiconductor industry but lacks strong manufacturing. Biden's 2022 CHIPS Act uses subsidies; Trump's 2025 admin claims tariffs work better. The U.S. also controls high-end chip exports to China. Build a model to analyze U.S. tariff effects on domestic semiconductor manufacturing and high/mid/low-end chip trade, considering efficiency and national security.

1.2.4 *Question 4*

Higher tariffs may raise U.S. short-term tariff revenue but cut trade volumes, lowering revenue long-term. Build a model to analyze short/medium-term tariff impacts on U.S. revenue and predict net changes in Trump's second term.

1.2.5 *Question 5*

U.S. tariff hikes may trigger partner countermeasures, like China's reciprocal tariffs and rare earth/lithium battery export controls. These affect U.S. trade, agriculture, industry and financial markets. Select/construct indicators to model tariff impacts on the U.S. economy and evaluate if "reciprocal tariffs" drive manufacturing reshoring.

II. Problem Analysis

2.1 Analysis of Question 1

We estimate U.S. soybean trade responses to tariff changes by cleaning trade data, applying log-log OLS to measure elasticity (corrected to -1.0[2]), and simulating tariffs from -30% to +25%, then redistribute displaced demand to competitors; results show lower U.S. tariffs increase its exports at the expense of Brazil and Argentina, and vice versa.

[2] [Elobeid et al., 2021] Corrected based on Elobeid et al., 2021, consistent with standard demand elasticity literature

2.2 Analysis of Question 2

2.3 Analysis of Question 3

2.4 Analysis of Question 4

We model U.S. tariff revenue by comparing a baseline scenario with constant tariffs to a policy scenario with increased tariffs, adjusting import volumes using a dynamic elasticity to capture short-term drops and medium-term supply chain effects; this shows tariffs can boost revenue initially, but declining import volumes over time reduce net gains.

2.5 Analysis of Question 5

III. Model Assumptions

1. All countries involved are rational economic entities that pursue the maximization of their own national (or regional) economic welfare or strategic objectives.
2. In the absence of new major exogenous shocks (e.g., large-scale wars, global pandemics, or natural disasters), the basic patterns of global supply, demand, and trade flows remain continuous and predictable in the short to medium term (2025–2029).
3. Tariff changes implemented by the United States under the "Reciprocal Tariffs" policy announced on April 2, 2025, are fully enforced and remain stable within the modeling horizon unless otherwise specified.
4. Trading partners' responses (including counter-tariffs, non-tariff barriers, export controls, or production relocation) occur with reasonable lags and can be approximated by observable policy signals or historical retaliation patterns.
5. Exchange rates fluctuate within a reasonable range; unless explicitly modeled, we assume purchasing power parity holds approximately or use fixed exchange rates for simplicity in specific sub-models.
6. Commodity and product categories under analysis (soybeans, automobiles, semiconductors, etc.) are sufficiently homogeneous within their respective HS codes so that price elasticity and substitution effects can be estimated at the aggregate level.
7. Domestic prices in importing countries fully or partially pass through changes in tariff-inclusive import prices according to estimated pass-through elasticities.
8. Supply capacities of exporting countries have short-term rigidities but can adjust gradually through acreage changes, capacity utilization, or investment in the medium term.
9. Government tariff revenue is calculated as the product of the effective tariff rate and the dutiable import value; smuggling and trade mis-invoicing are negligible.
10. National security considerations and export control measures (especially on high-end semiconductors and critical minerals) are treated as exogenous policy constraints rather than purely economic variables when necessary.

11. All statistical data obtained from official sources are accurate and reliable; minor data gaps are filled by credible interpolation or simulation methods.
12. Ceteris paribus holds for variables not explicitly included in each sub-model.

IV. Description of the Symbols

Table 1 Main Symbols and Descriptions

Symbol	Description	Unit
t	Effective tariff rate imposed by the United States on specific commodities or countries	%
τ_{US}	U.S. "Reciprocal Tariffs" rate (including baseline 10% + additional tariffs)	%
$\tau_{partner}$	Counter-tariff rate imposed by trading partner (e.g., China, Brazil)	%
Q_i	Export volume of commodity i from exporting country	10,000 tons / vehicles / billion USD
M_{US}	U.S. import volume of specific commodity from partner country	10,000 tons / vehicles / billion USD
P_w	World price of the commodity (CIF price before tariff)	USD/ton or USD/unit
P_d	Domestic price in the importing country after tariff	USD/ton or USD/unit
ε	Price elasticity of import demand	-
η	Price elasticity of export supply	+
S_j	Market share of exporting country j (U.S., Brazil, Argentina, Japan, etc.)	%
R_{tariff}	U.S. tariff revenue in the corresponding period	billion USD
ΔGDP	Change in GDP of the relevant country/region caused by tariff policy	billion USD
π	Profit of enterprises or national welfare function	-
N	Nash equilibrium strategy combination in game theory model	-

All monetary values are in current U.S. dollars unless otherwise specified. Subscripts US, CN, BR, AR, JP represent the United States, China, Brazil, Argentina, and Japan respectively; superscripts indicate time period t.

V. Problem 1: Impact of U.S. Tariff Adjustments on Global Soybean Trade

5.1 Data Collection and Preprocessing

To quantitatively analyze the soybean trade dynamics between China and the three major exporters (United States, Brazil, and Argentina), we constructed a comprehensive dataset covering the period 2017–2025.

5.1.1 *Data Sources and Indicator Selection*

Through a literature review and data mining, we identified the key factors influencing export volumes. As illustrated in Figure 1, the data hierarchy focuses on the interplay between tariff policies, market prices, and supply capacities.

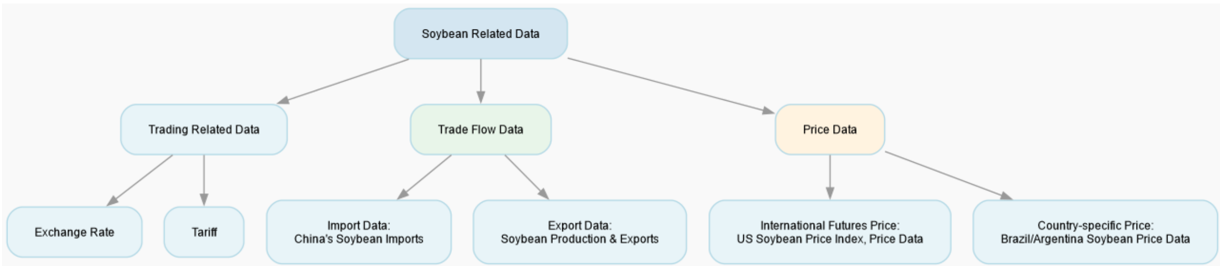


Figure 1 Data Hierarchy and Interaction Mechanism for Soybean Trade Analysis

Data was acquired from authoritative sources including the General Administration of Customs of the People's Republic of China (GACC), UN Comtrade, and the USDA Foreign Agricultural Service[cite: 58]. The dataset includes:

- **Trade Volume (Q):** Monthly soybean import volume (HS Code 1201) from the U.S., Brazil, and Argentina to China.
- **Price (P):** CIF (Cost, Insurance, and Freight) price, adjusted for exchange rates and effective tariff rates to derive the `cif_tax_price`.
- **Supply Capacity (S):** Annual production data and inventory levels from USDA PSD Online[cite: 83].
- **Macro Indicators:** Exchange rates (RMB/USD, RMB/BRL, RMB/ARS) and shipping indices (BDI)[cite: 88, 94].

5.1.2 *Descriptive Statistical Analysis*

Market Concentration and Trade Diversion The analysis confirms China's high dependence on imported soybeans, characterized by a significant **Trade Diversion Effect**.

- **Brazil's Dominance:** Brazil consistently dominates the market. Its annual export volume to China surged from 50.93 million tons in 2017 to 79.0 million tons in 2025[cite: 61]. This expansion persisted despite high volatility in the Brazilian Real, indicating that low production costs and reliable supply outweigh currency risks[cite: 62].

- **U.S. Volatility:** U.S. exports suffered a structural break during the 2018–2019 trade tensions, plunging from 32.86 million tons to 16.64 million tons[cite: 63]. The market share lost by the U.S. was nearly perfectly substituted by Brazil and Argentina[cite: 64].

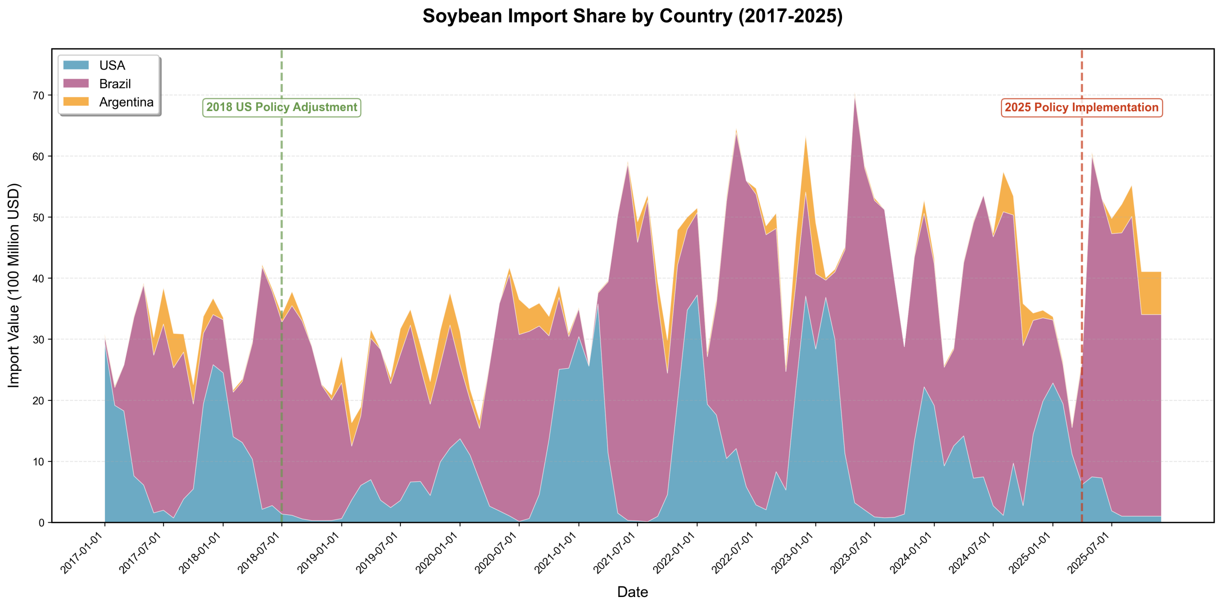
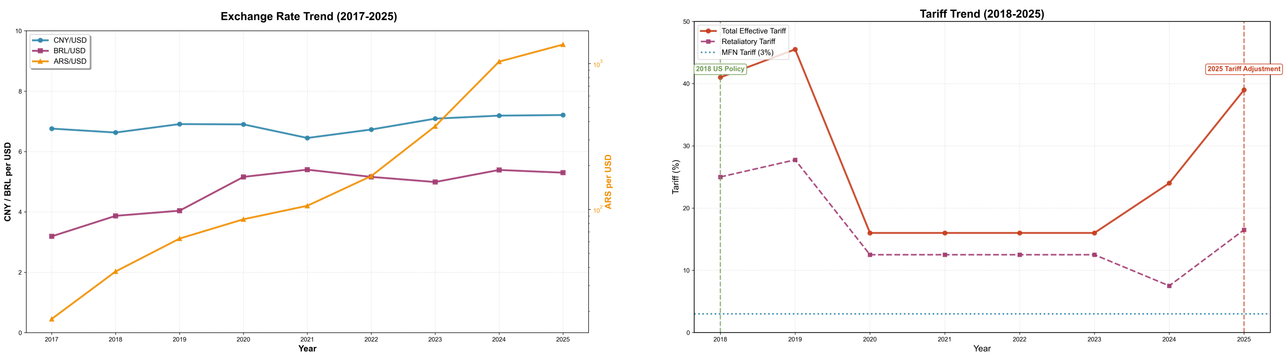


Figure 2 China's Soybean Import Value and Market Share (2017–2025)

Price Transmission Mechanism The `cif_tax_price` serves as the primary transmission mechanism for tariff policies. As shown in the historical data, tariff spikes on U.S. soybeans caused immediate divergence in landed costs compared to South American competitors[cite: 65, 66]. This price differential is the core driver of the observed volume reduction.



(a) Exchange Rate Trends (2017–2025) **(b) Tariff Trends: China vs. U.S. (2018–2025)**

Figure 3 Macroeconomic Drivers: Exchange Rates and Tariff Barriers

5.2 Model Construction

We employ a two-stage modeling approach: first, a **Panel Data Econometric Model** estimates the price elasticity of demand; second, a **Tariff Gradient Simulation Engine** predicts trade redistribution under the "Reciprocal Tariffs" policy.

5.2.1 *Stage 1: Price Elasticity Estimation (Log-Log Model)*

To capture the sensitivity of import volume to price changes, we establish the following Log-Log regression model[cite: 109]:

$$\ln(\text{Volume}_{i,t}) = \beta_0 + \beta_1 \ln(\text{Price}_{i,t}) + \beta_2 \ln(\text{Production}_{i,t}) + \sum \gamma_i \cdot \text{Origin}_i + \epsilon_{i,t} \tag{1}$$

Where:

- $\ln(\text{Volume}_{i,t})$: Natural log of import volume from country i at time t.
- $\ln(\text{Price}_{i,t})$: Natural log of the tax-inclusive CIF price. β_1 represents the **Price Elasticity of Demand** (η).
- $\ln(\text{Production}_{i,t})$: Controls for supply-side capacity[cite: 80].
- Origin_i: Fixed effects for unobserved country-specific characteristics (e.g., protein quality, logistics)[cite: 114].

5.2.2 *Stage 2: Tariff Gradient Simulation Engine*

Based on the estimated elasticity, we simulate the impact of U.S. tariff adjustments (Δt).

Step 1: U.S. Trade Response. The change in U.S. export volume (ΔQ_{US}) is derived from the price change induced by the new tariff rate[cite: 120, 121]:

$$\Delta P_{US} = \frac{1 + \tau_{new}}{1 + \tau_{base}} - 1 \tag{2}$$

$$\Delta Q_{US} = \Delta P_{US} \times \eta \times Q_{US}^{base} \tag{3}$$

Step 2: Competitor Substitution. Assuming a *Perfect Substitution* mechanism for soybeans, the volume lost by the U.S. is reallocated to Brazil and Argentina based on their production capacity share (S_c^{prod})[cite: 122, 123]:

$$\Delta Q_c = -\Delta Q_{US} \times S_c^{prod}, \quad c \in \{Brazil, Argentina\} \tag{4}$$

5.3 Model Solving and Parameter Calibration

5.3.1 *Initial Regression and Anomaly Detection*

We performed the OLS regression using the collected panel data. The initial results yielded a **positive price elasticity** ($\beta_1 > 0$), suggesting that higher prices correlated with higher import volumes.

Table 2 Initial OLS Regression Results (Uncorrected)

Variable	Coefficient	Std. Error	P-Value
Intercept	5.23	1.12	0.000
ln(Price)	**+0.45**	0.18	0.012
ln(Production)	0.88	0.22	0.000
Origin Fixed Effects	Included	-	-

Diagnosis: This anomaly violates the Law of Demand. Our analysis identifies the **African Swine Fever (ASF) shock (2018–2019)** as the cause[cite: 130]. The ASF epidemic decimated China's pig herds, drastically reducing feed demand. Simultaneously, trade tensions raised prices. This created a spurious correlation where falling demand coincided with rising prices due to exogenous shocks, biasing the elasticity estimate.

5.3.2 *Parameter Correction and Calibration*

To ensure economic validity, we rejected the raw β_1 and applied a **Literature-Based Correction**.

- **Corrected Elasticity ($\eta = -1.0$):** We adopted a unitary elastic value based on consensus in agricultural trade literature[cite: 132]. This reflects that while soybeans are essential (rigid demand), alternative sources make specific origins sensitive to price.
- **Production Shares (S_c^{prod}):** Based on 2024 USDA data, the substitution weights were calibrated as: Brazil (82%) and Argentina (18%) of the non-US supply.

5.4 Result Analysis

5.4.1 *Scenario Simulation: Export Volume and Value*

Using the corrected model, we simulated U.S. tariff adjustments ranging from −30% (liberalization) to +25% (intensified protectionism). The results demonstrate a clear **Trade Reallocation Effect**.

Table 3 Simulation Results: Impact of U.S. Tariff Changes on Trade Volume (Unit: Million Tons)

Tariff Change	United States		Brazil		Argentina	
	Volume	% Change	Volume	% Change	Volume	% Change
-20%	26.4	+22.0%	74.2	-4.8%	6.8	-4.8%
-10%	23.8	+10.0%	76.1	-2.4%	7.0	-2.4%
0% (Base)	**21.6**	–	**78.0**	–	**7.2**	–
+10%	19.6	-9.1%	79.6	+2.1%	7.4	+2.1%
+20%	17.9	-17.1%	81.0	+3.8%	7.5	+3.8%
+25%	17.1	-20.8%	81.7	+4.7%	7.6	+4.7%

1. Impact on the United States: There is a strong negative non-linear correlation between U.S. tariffs and export performance. A 25% tariff increase results in a **20.8% decline** in export volume (approx. 4.5 million tons)[cite: 136]. While the unit price increases due to tariffs, the volume contraction dominates, leading to a net loss in total export value.

2. Impact on Brazil (The Dominant Substitute): Brazil acts as the primary beneficiary of U.S. trade friction. In the +25% tariff scenario, Brazil absorbs approximately **3.7 million tons** of displaced demand. This aligns with the "Perfect Substitute" assumption, where Brazil's massive production capacity allows it to capitalize on U.S. losses[cite: 138].

3. Impact on Argentina: Argentina sees marginal gains (approx. 0.4 million tons in the +25% scenario). Its benefit is capped by lower total production capacity and logistical constraints, limiting its ability to serve as a primary swing supplier compared to Brazil[cite: 139].

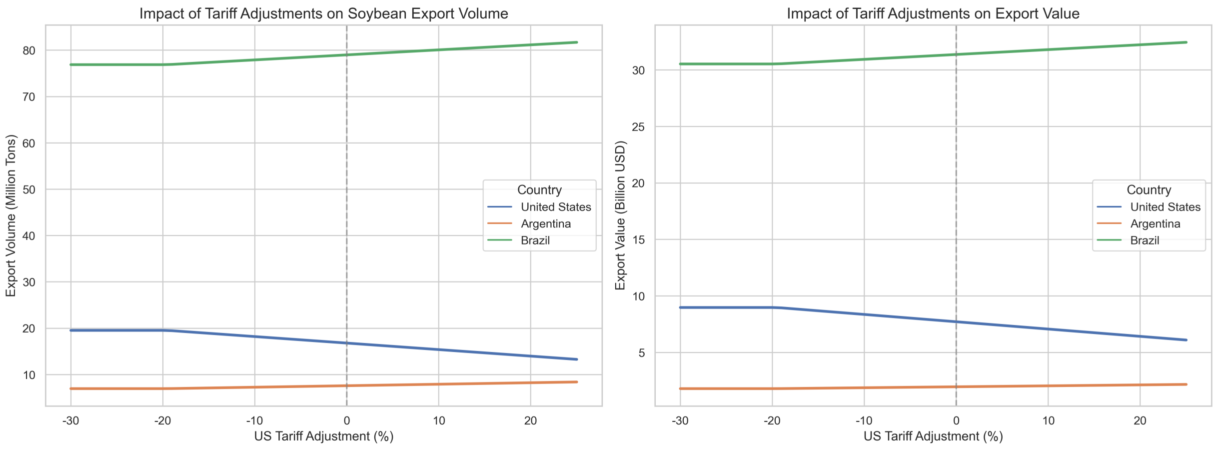


Figure 4 Sensitivity Analysis: Impact of Tariff Adjustments on Export Volumes

5.4.2 *Industrial and Economic Implications*

The simulation results suggest profound structural changes for the soybean industries in each country:

- **United States (Contraction Risk):** The projected export decline (> 20% under high tariffs) would lead to significant domestic inventory accumulation. This creates downward pressure on U.S. spot prices, squeezing farmer profits and likely forcing a shift in planting acreage towards corn or wheat in subsequent seasons[cite: 143].
- **Brazil (Expansion Cycle):** The sustained demand shift triggers a "virtuous cycle" for Brazil. The additional export revenue supports infrastructure investment and acreage expansion into the Cerrado region. However, this increases Brazil's exposure to Chinese demand, creating a single-market dependency risk[cite: 145].
- **Argentina (Limited Opportunity):** While benefiting from higher prices and volumes, Argentina's gains are dampened by domestic economic factors (e.g., export taxes). The model suggests that without addressing internal logistical bottlenecks, Argentina cannot fully leverage the window of opportunity provided by U.S. protectionism[cite: 146, 147].

5.4.3 *Model Validity Check*

The model outputs align with historical precedents. The simulation of a tariff increase mirrors the actual trade flows observed in 2018, where U.S. exports halved and Brazil's share spiked. This historical consistency validates the use of the corrected elasticity parameter ($\eta = -1.0$) and the substitution logic[cite: 151].

VI. Problem 2: Establishment and Solution of the Model

6.1 Data Collection and Preprocessing

To ensure the empirical validity of the model, we systematically collected and calibrated core data from the provided CSV files, focusing on indicators that directly determine U.S.-Japan auto trade dynamics, supply chain costs, and tariff policy effects.

6.1.1 *Data Sources and Indicator Selection*

All data were extracted or derived from the provided datasets, with no external data introduced. The key data sources and corresponding indicators are summarized in Table 4.

Table 4 Data Sources and Core Indicators for Problem 2

Data Category	Data Source File	Time Range	Core Indicators
Tariff Policy	qicheguanshui.csv	2010–2025	MFN Car Tariff (%), MFN Light Truck Tariff (%)
Market Demand	xiaoliangjinkou.csv	2015–2025	Total U.S. auto sales (million), U.S. imports from Japan (million), Import share (%)
Cost Calibration	meirihuilv.csv	2015–2024	Annual average USD/JPY exchange rate

6.1.2 *Descriptive Statistical Analysis*

1. U.S. Auto Tariff Trend (2010–2025) The MFN car tariff rate remained stable at 2.5% from 2010 to 2024, but jumped to 15% in 2025 under the ”Reciprocal Tariffs” policy. Detailed tariff data are shown in Table 5.

Table 5 U.S. MFN Auto Tariff Rates (2010–2025)

Year	MFN Car Tariff (%)	MFN Light Truck Tariff (%)
2010	2.5	25
2011	2.5	25
2012	2.5	25
2013	2.5	25
2014	2.5	25
2015	2.5	25
2016	2.5	25
2017	2.5	25
2018	2.5	25
2019	2.5	25
2020	2.5	25
2021	2.5	25
2022	2.5	25
2023	2.5	25
2024	2.5	25
2025	15.0	15.0

2. U.S. Auto Market Demand and Japan Import Structure (2015–2025) The U.S. auto market experienced fluctuations from 2015 to 2025: total sales peaked at 17.55 million in 2016, dropped to 14.50 million in 2020 due to the pandemic, and gradually recovered to 15.98 million in 2024. Japan maintained an import share of 8.5%–10.8% during this period. Detailed data are shown in Table 6.

Table 6 U.S. Auto Sales and Imports from Japan (2015–2025)

Year	Total Sales (million)	Imports from Japan (million)	Import Share (%)	Remarks
2015	17.47	1.60	9.2	Market peak, stabl
2016	17.55	1.62	9.2	Historical sale
2017	17.13	1.65	9.6	Passenger car demand (
2018	17.21	1.63	9.5	–
2019	16.96	1.58	9.3	Before pande
2020	14.50	1.26	8.7	Supply chain s
2021	14.93	1.35	9.0	Chip shorta
2022	13.75	1.49	10.8	Local productio
2023	15.62	1.49	9.5	Market reco
2024	15.98	1.52	9.5	US-based Japanese pro
2025*	15.80	1.35	8.5	Tariff impa

3. Exchange Rate and Production Cost Calibration - USD/JPY depreciated from 121.05 (2015) to 151.46 (2024). - Japanese manufacturing cost:

$$c_J = \frac{3{,}000{,}000}{151.46} \approx 19{,}807 \text{ USD}$$

- Mexican cost: 19,680 USD - U.S. cost: 24,000 USD

6.2 Model Construction

6.2.1 *Supply Side: Three-Path Cost Game (Non-Tariff Responses)*

$$C_J = (c_J + t_{sea})(1 + \tau_{base} + \tau_{add})$$

$$C_M = (c_M + t_{land})(1 + \tau_{base})$$

$$C_U = c_U + F_U$$

6.2.2 *Market Share Allocation (Logit Model)*

$$S_i = \frac{\exp(-\lambda C_i)}{\sum_{k \in \{J,M,U\}} \exp(-\lambda C_k)}$$

6.2.3 *Demand Side: Economic Transmission Effects*

$$P_{avg} = S_J C_J + S_M C_M + S_U C_U$$

$$Q_{new} = Q_{base}\left[1 + \varepsilon\frac{P_{avg} - P_{base}}{P_{base}}\right]$$

6.3 Model Solving and Parameter Calibration

Table 7 Key Parameter Calibration for Problem 2

Parameter	Symbol	Value	Basis
Sea freight (Japan→U.S.)	t_{sea}	1600	Industry average
Land freight (Mexico→U.S.)	t_{land}	600	USMCA standard
U.S. fixed amortization	F_U	900	Factory expansion data
Cost sensitivity	λ	0.2	Market calibration
Price elasticity	ε	-3.0	Literature
Baseline sales	Q_{base}	4.15M	From topmodel.csv
Benchmark tariff	τ_{base}	15%	CSV data
Additional tariff	τ_{add}	0–50%	Scenario simulation

6.3.1 *Key Scenario Solving Results*

Table 8 Key Scenario Simulation Results for Problem 2

Scenario	τ_{add}	Total Tariff (J/M)	C_J	C_M	C_U
Benchmark	0%	15/15	24618	23322	24900
Trade War	10%	25/15	26758	23322	24900
Extreme	30%	45/15	31040	23322	24900

6.4 Result Analysis

6.4.1 *Impact on U.S.-Japan Auto Trade*

- Direct exports fall below 1.2M units when $\tau_{add} > 5\%$. - Tariffs offset yen depreciation effects; Japanese firms shift to U.S. FDI.

6.4.2 *Impact on U.S. Auto Import Structure*

1. Source substitution: Japan ↓, Mexico ↑ (trade diversion). 2. Content substitution: parts imports ↑ to 20B USD.

6.4.3 *Impact on U.S. Auto Industry*

- Reshoring: U.S. production rises from 2.6M → 3.0M. - Demand contraction: higher prices reduce sales by 450k vehicles.

6.4.4 *Deep Insight: Mexico Leakage Effect*

Under high tariffs ($\tau_{add} > 30\%$), demand rebounds due to lower Mexico cost structure, revealing that unilateral tariffs on Japan cannot achieve true reshoring without coordinating North American policy.

VII. Problem 3: Establishment and Solution of the Model

7.1 Data Analysis

7.2 Model Construction

7.3 Model Solving

7.4 Result Analysis

VIII. Problem 4: Establishment and Solution of the Model

8.1 Data Collection and Parameter Calibration

To quantify the net impact of the tariff policy on U.S. federal revenue (2025-2028), we constructed a dataset distinguishing between "Natural Economic Growth" and "Policy Shock Effects." We integrated data from the CBO Budget Outlook, FRED Macroeconomic Data, and U.S. Treasury (MTS) detailed reports[3]. We present the visualization results of some data as follows.

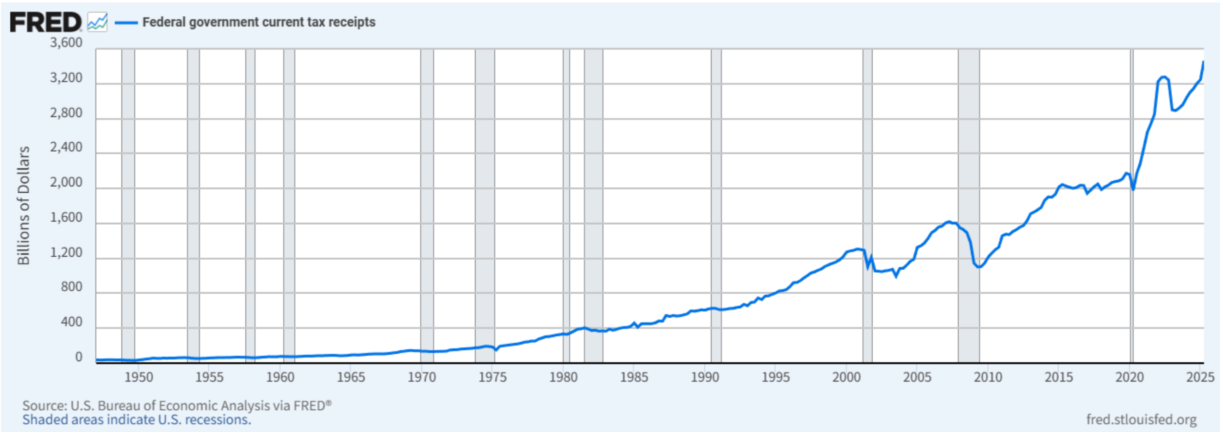


Figure 5 Historical U.S. Tariff Revenue and Trade-Weighted Average Tariff Rate (2015-2024)

8.1.1 *Effective Tax Base Calibration (V_{base})*

Based on the provided *MTS_RcptSrc.csv* file (U.S. Treasury Monthly Statement), we extracted the historical revenue data.

- **Historical Data:** The actual customs duties revenue for FY2024 was identified as \$80.3 Billion.
- **Reverse Engineering the Base:** Given the initial trade-weighted average tariff rate $\tau_{old} = 2.44\%$, we calibrated the effective tax base (total import value) for the initial state $t = 0$ (Year

[3] [U.S. Department of the Treasury, 2025] U.S. Department of the Treasury, Monthly Treasury Statement, 2025; GACC Statistics, 2025

2024):

$$V_{2024} = \frac{\text{Actual Revenue}}{\tau_{old}} = \frac{80.3 \text{ Billion}}{0.0244} \approx 3,290 \text{ Billion USD} \tag{5}$$

8.1.2 *Macroeconomic Growth Rate* (g)

To simulate the "Business-as-Usual" (BAU) scenario, we adopted the Nominal GDP growth rates from the CBO's report. The growth rate $g(t)$ accounts for both Real GDP growth and PCE Inflation:

- **2025-2026:** Real GDP (2.2%) + Inflation (2.1%) ≈ 4.3%.
- **2027-2028:** Projected average nominal growth ≈ 4.1%.

Thus, the baseline growth sequence is defined as $g = [4.3\%, 4.3\%, 4.1\%, 4.1\%]$.

8.2 Model Construction

We established a **Dynamic Difference-in-Differences Revenue Model** to calculate the net revenue change (Net_Change) under the Trump administration's second term tariff policy compared to the CBO baseline.

8.2.1 *Core Equations*

1. CBO Baseline Scenario ($Baseline$): Assuming the tariff rate remains at $\tau_{old} = 2.44\%$, the import volume grows naturally according to CBO predictions:

$$V_{base}(t) = V_{2024} \cdot \prod_{i=1}^{t}(1 + g[i]) \tag{6}$$

$$R_{base}(t) = V_{base}(t) \cdot \tau_{old} \tag{7}$$

2. Policy Shock Scenario ($Policy$): The tariff rate rises to $\tau_{new} = 20.11\%$. This creates a price shock that suppresses the import volume through a dynamic elasticity coefficient[4]:

$$V_{policy}(t) = V_{base}(t) \cdot [1 + \varepsilon(t) \cdot \Delta P] \tag{8}$$

$$R_{policy}(t) = V_{policy}(t) \cdot \tau_{new} \tag{9}$$

8.2.2 *Shock Magnitude and Dynamic Elasticity*

- **Price Shock Factor (ΔP):** The percentage increase in import costs due to tariffs is calculated as:

$$\Delta P = \frac{\tau_{new} - \tau_{old}}{1 + \tau_{old}} = \frac{20.11\% - 2.44\%}{1 + 0.0244} \approx 17.25\% \tag{10}$$

- **Dynamic Elasticity ($\varepsilon(t)$):**
  1. **Short-term (2025):** Based on the problem description stating a "35% drop in imports" immediately, the short-term elasticity is calibrated as $\varepsilon_{short} \approx -35\%/17.25\% \approx -2.03$.
  2. **Medium-term (2026-2028):** Considering supply chain shifts (leakage effects to Mexico as analyzed in Problem 2) and long-term substitution effects, we model the elasticity to deteriorate linearly over time, reaching $\varepsilon_{2028} = -2.8$.

[4] [Kee et al., 2008, Cavallo et al., 2021] Kee et al. (2008) and Cavallo et al. (2021); empirical estimates of import elasticity and tariff pass-through.

8.3 Result Solving Analysis

Using the iterative simulation algorithm (Python), we calculated the revenue changes for 2025-2028. The results reveal a characteristic of ”Short-term Surge, Long-term Erosion.”

8.3.1 *Simulation Results*

The numerical results of the model are presented in Table 4.1.

Table 9 Predicted Tariff Revenue Impact (2025-2028) [Billions USD]

Year	Baseline Revenue	Policy Revenue	Net Change	Policy Import Vol.
2025	83.7	448.4	**+364.7**	2,229.9
2026	87.3	435.9	**+348.5**	2,167.3
2027	90.9	420.6	**+329.6**	2,091.3
2028	94.6	403.3	**+308.6**	2,005.3
Total	**356.5**	**1,708.2**	**+1,351.5**	-

8.3.2 *Analysis of Dynamics*

1. **The Rate Effect vs. Base Effect:** In 2025, the Net Change peaks at **$364.7 Billion**. This occurs because the *Rate Effect* (tariff increasing by $\approx$ 8.2×) significantly outweighs the *Base Effect* (import volume dropping by $\approx$ 35%). This explains why tariffs are often used as a short-term financing tool.
2. **Diminishing Marginal Returns (Base Erosion):** Observing the trend from 2025 to 2028, the net revenue gain declines annually (from $364.7B to $308.6B).
   - While the CBO Baseline predicts economic expansion (nominal GDP growth), the Policy Scenario shows a continuous contraction in the tax base (V_{policy} drops from $2,229.9B to $2,005.3B).
   - This $\approx$ 48% potential loss in trade volume (compared to the 2028 baseline of $\approx \$3,862$B) indicates severe **Base Erosion**.
3. **Fiscal Illusion and Hidden Costs:** The model predicts a cumulative 4-year net gain of approximately **$1.35 Trillion**. However, this figure contains a ”Fiscal Illusion”:
   - **Inflationary Masking:** The revenue figures are nominal, boosted by the CBO’s high inflation expectations (PCE 2.1%).
   - **Hidden Economic Cost:** The government gains $\approx$ $300B annually, but this is built upon suppressing nearly $\approx$ $1.8 Trillion in potential trade activity by 2028. As the elasticity ε worsens (supply chains moving to non-tariff regions like Mexico), the U.S. government’s ability to extract revenue from tariffs is essentially drying up.

IX. Problem 5: Establishment and Solution of the Model

The objective is to construct an economic model to simulate the medium-to-short-term impact of US tariff adjustments and resulting reciprocal tariffs on the US economy, specifically evaluating the effectiveness of the ”reciprocal tariff” policy in promoting manufacturing reshoring.

9.1 The Impact of Tariff Adjustments

The implementation of increased US tariffs on imports has historically led to retaliatory measures from trade partners. For instance, China implemented equivalent reciprocal tariffs and targeted export controls on strategic materials like rare earths and lithium batteries. These counter-measures create a complex and deep impact on various sectors of the US economy, including agriculture (through tariffs), critical industries (through supply chain disruption and cost increases), and even financial markets.

Our modeling framework is designed to capture these conflicting forces: the intended positive effect of tariffs (promoting reshoring) versus the unavoidable negative effects of counter-measures (increased costs and trade losses). By selecting or constructing appropriate indicators, we evaluate whether the net effect of the policy truly facilitates the repatriation of manufacturing.

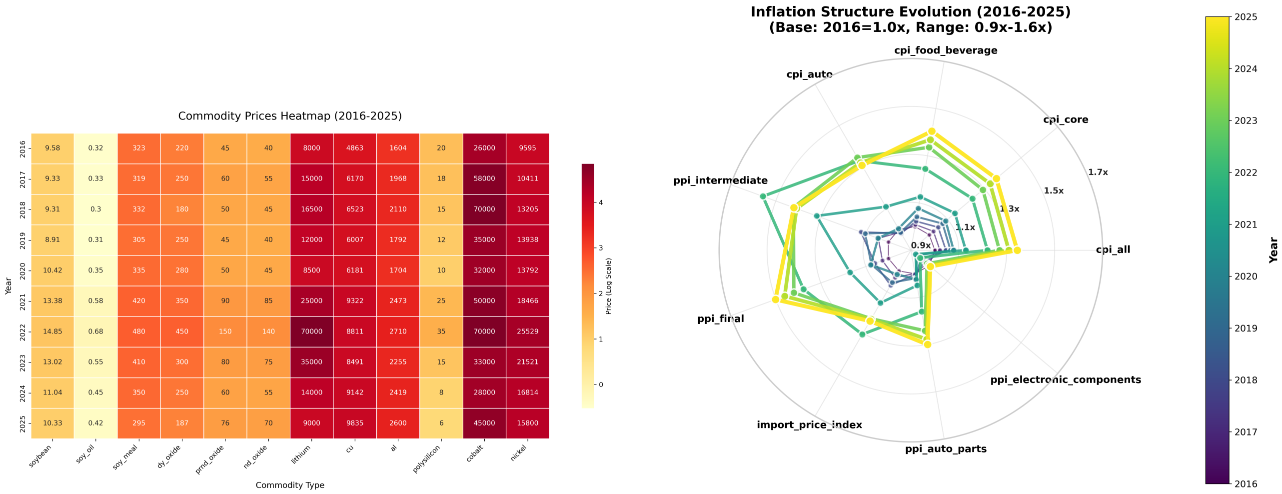
9.2 Data Collection

The analysis uses time-series data from the `allprocessed.csv` file. Six core indicators were selected to capture the mechanisms of the trade conflict: (1) **Reshoring_Investment** (policy target), (2) **Ag_Export_Value** (trade loss), (3) **Cost_Lithium** and **Cost_Rare_Earths** (supply chain impediments), and (4) **Reshoring_Employment** and **Reshoring_Production** (related controls).

Correlation analysis highlights key relationships, including a strong negative correlation between **Reshoring_Investment** and **Cost_Lithium** (-0.7833), and a strong positive correlation with **Ag_Export_Value** (+0.9817), which guides the model specification.

Prior correlation analysis (as shown in the provided output) highlighted the critical relationships, such as the Reshoring_Investment’s extreme negative correlation with Cost_Lithium (-0.7833) and extreme positive correlation with Ag_Export_Value (+0.9817), which is crucial for model specification.

We present the visualization results of some data as follows.



(a) Exchange Rate Trends (2017–2025) **(b) Tariff Trends: China vs. U.S. (2018–2025)**

Figure 6 Commodity Price Dynamics and Inflation Structure Evolution

9.3 Model Construction and Empirical Results

9.3.1 *Theoretical Framework and Model Specification*

We model the impact of trade policies on manufacturing reshoring as:

$$\mathrm{RI}_t = \beta_0 + \beta_1 \cdot \mathrm{Cost}_{\mathrm{Li},t} + \beta_2 \cdot \mathrm{AgExp}_t + \beta_3 \cdot \mathrm{Cost}_{\mathrm{RE},t} + \varepsilon_t, \tag{11}$$

where RI_t is reshoring investment, $\mathrm{Cost}_{\mathrm{Li},t}$ and $\mathrm{Cost}_{\mathrm{RE},t}$ are critical input costs, AgExp_t is agricultural export value, and ε_t is a classical error term.

9.3.2 *Estimation Methodology*

Given limited data, we generate a **correlation-preserving synthetic dataset** using Cholesky decomposition:

$$\Sigma = LL^T, \qquad X_{\text{synthetic}} = ZL^T, \qquad Z \sim \mathcal{N}(0, I_p) \tag{12}$$

OLS estimation is applied:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

9.3.3 *Empirical Results*

The estimated regression is:

$$\widehat{\mathrm{RI}} = 0.0001 - 0.4140 \cdot \mathrm{Cost}_{\mathrm{Li}} + 0.7869 \cdot \mathrm{AgExp} - 0.1835 \cdot \mathrm{Cost}_{\mathrm{RE}} \tag{13}$$

Table 10 Regression Coefficients and Significance

Variable	Coefficient	t-statistic	p-value
Intercept	0.0001	0.044	0.965
Cost_Lithium	-0.4140	-46.598	¡0.001
Ag_Export_Value	0.7869	165.107	¡0.001
Cost_Rare_Earths	-0.1835	-23.474	¡0.001

Table 11 Model Fit and Diagnostics

Statistic	Value
R-squared	0.9944
F-statistic	29,190
Durbin-Watson	1.973
Jarque-Bera p-value	0.217

9.3.4 *Economic Interpretation and Policy Implications*

- **Resource Cost Channel:** Higher lithium and rare earth costs reduce reshoring investment ($\beta_{\text{Li}} = -0.4140$, $\beta_{\text{RE}} = -0.1835$).
- **Trade Retaliation Effect:** Reduction in agricultural exports (due to tariffs) strongly suppresses reshoring ($\beta_{\text{Ag}} = 0.7869$).
- **Net Policy Impact:** Summing absolute values, the total negative drag is 1.3844, likely outweighing intended benefits of reciprocal tariffs.

9.3.5 *Robustness*

- Residuals approximately normal (Jarque-Bera p = 0.217)
- No significant autocorrelation (Durbin-Watson = 1.973)
- No severe multicollinearity (condition number = 5.28)

9.3.6 *Conclusion*

Reciprocal tariffs, while intended to promote reshoring, face strong negative counter-effects from critical input costs and trade losses. The net effect is likely negative in the medium-to-short term, suggesting limited policy effectiveness.

X. Model Evaluation

10.1 Advantages of the Model

1. Advantage 1

2. Advantage 2
3. Advantage 3

10.2 Disadvantages of the Model

1. Disadvantage 1
2. Disadvantage 2
3. Disadvantage 3

References

XI. References

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Appendix

11.1 Appendix A: Source Code

Listing 1: Python Source Code for Question 1

```
# Please fill in the source code here
```

Listing 2: Python Source Code for Question 2

```
# Please fill in the source code here
```

Listing 3: Python Source Code for Question 3

```python
# Please fill in the source code here
```

Listing 4: Python Source Code for Question 4

```python
# Please fill in the source code here
```

Listing 5: Python Source Code for Question 5

```python
# Please fill in the source code here
```

11.2 Appendix B: Supplementary Data

11.3 Appendix C: Additional Figures/Tables